



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

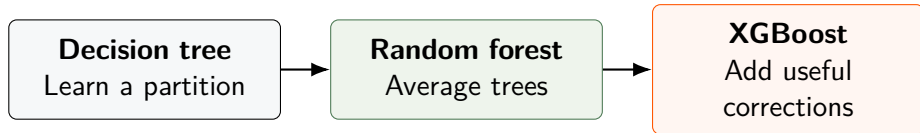
COMP 5212

Machine Learning

Instructor: Zihan Zhang

Lecture 7: Decision Tree, Random Forest and XGboost

Learning Goals

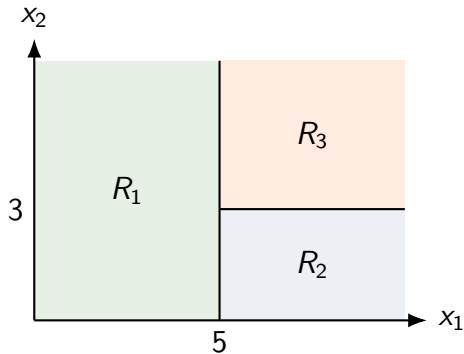
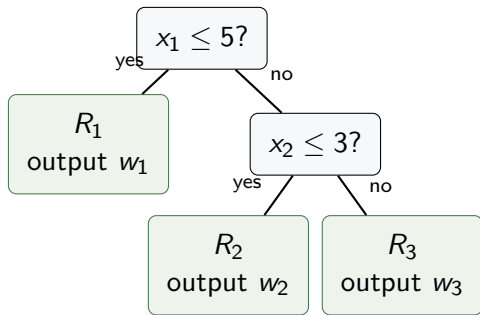


By the end of the lecture, you should be able to:

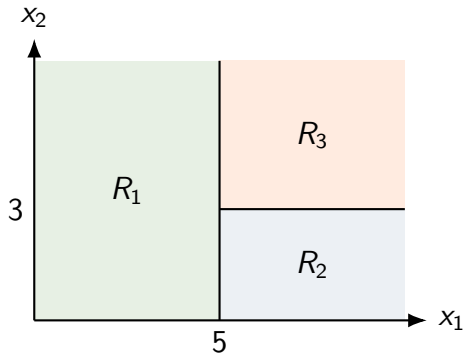
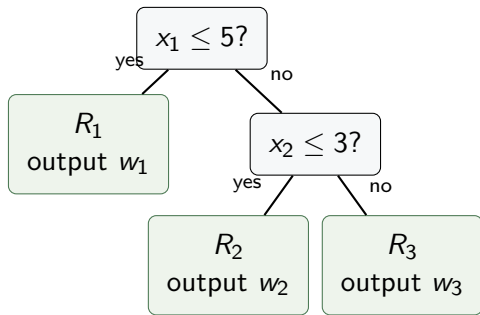
- Compare split criteria and compute leaf predictions;
- Explain feature randomization and out-of-bag evaluation;
- Compare random forests and XGboost; understand the pros and cons

Decision Trees

A Tree Partitions the Feature Space



A Tree Partitions the Feature Space



$$f(x) = \sum_{v=1}^J w_v \mathbf{1}\{x \in R_v\}.$$

Each leaf defines a region with a constant prediction.

What Does a Leaf Predict?

Regression

For squared error, use the mean target in the leaf:

$$w_v = \frac{1}{|I_v|} \sum_{i \in I_v} y_i.$$

Targets: 2, 4, 6

Prediction: 4

Classification

Estimate class proportions in the leaf:

$$p_v(c) = \frac{\sum_{i \in I_v} \mathbf{1}\{y_i = c\}}{|I_v|}.$$

Labels: 0, 1, 1, 1

$p_v(1) = 0.75$;

predict class 1

I_v : training samples reaching leaf v ; J : number of leaves.

Choose a Criterion, Then Compare Splits

Split the samples I into children I_L and I_R .

Let $A(I)$ be node impurity or average prediction loss.

$$\Delta = A(I) - \frac{|I_L|}{|I|}A(I_L) - \frac{|I_R|}{|I|}A(I_R).$$

Select the allowed split with the largest decrease.

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Regression

Squared error: use a leaf mean.

Absolute error: use a leaf median.

Classification

Gini impurity or entropy.

Prefer purer child nodes.

Weight each child by its fraction of the samples. The criterion is a modeling choice.

Regression: Minimize Squared Error

SSE: sum of squared errors. For a leaf containing samples I ,

$$\text{SSE}(I) = \sum_{i \in I} (y_i - \bar{y}_I)^2, \quad \text{MSE}(I) = \frac{\text{SSE}(I)}{|I|}.$$

Feature x	1	2	3	4
Target y	1	2	8	9

One leaf

$$\bar{y} = 5, \quad \text{SSE} = 50.$$

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$$\Delta_{\text{MSE}} = \frac{\text{SSE}(I) - \text{SSE}(I_L) - \text{SSE}(I_R)}{|I|}.$$

Classification: Gini or Entropy

Let p_c be the fraction of samples in class c at the node.

Gini impurity

$$G(I) = 1 - \sum_{c=1}^C p_c^2.$$

Measures class mixing.

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$$H(I) = - \sum_{c=1}^C p_c \log_2 p_c.$$

Label uncertainty in bits; $0 \log_2 0 = 0$.

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Binary class proportions	Gini	Entropy
Pure: (1, 0)	0	0
Balanced: (0.5, 0.5)	0.5	1

Both favor pure nodes, but may choose different splits.

One Split, Two Impurity Measures

Node	Class 0	Class 1	Gini G	Entropy H
Parent	4	4	0.500	1.0000
Left child	3	1	0.375	0.8113
Right child	1	3	0.375	0.8113

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$$\Delta_G = 0.5 - \frac{4}{8}(0.375) - \frac{4}{8}(0.375) = 0.125.$$

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Entropy decrease: information gain

$$\Delta_H = 1 - \frac{4}{8}(0.8113) - \frac{4}{8}(0.8113) \approx 0.1887 \text{ bits.}$$

Gain scales differ: compare candidates within one criterion.

Grow the Tree One Split at a Time

1. Start with all training samples at the root.
2. At a node, score candidate feature–threshold pairs using the chosen criterion.

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Local search

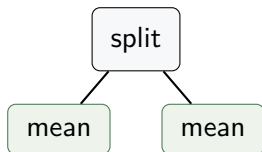
An early split changes every later decision in its subtree.

Computational shortcut

Sort a numerical feature and scan candidate cut points.

Greedy splitting does not guarantee the globally best tree.

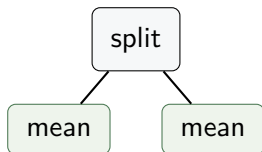
Control How Much the Tree Can Fit



Few broad regions

- Limit maximum depth.
- Require enough samples in each leaf.
- Prune a larger tree using a complexity penalty.

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Pruning objective: $R(T) + \alpha |\text{leaves}(T)|$.

$R(T)$: total training loss or impurity across leaves.

Choose complexity on validation data; deeper trees can have higher variance.

Learn Where Missing Values Should Go

Using Gini impurity, compare both missing-value directions for $x \leq 5$.

x	1	2	8	9	missing	missing
y	0	0	1	1	1	1

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Missing direction	Left labels	Right labels	Weighted Gini
Left	0, 0, 1, 1	1, 1	1/3
Right	0, 0	1, 1, 1, 1	0

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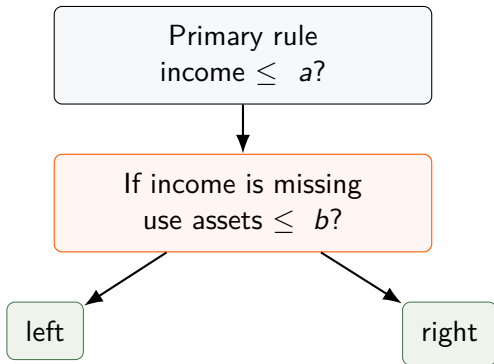
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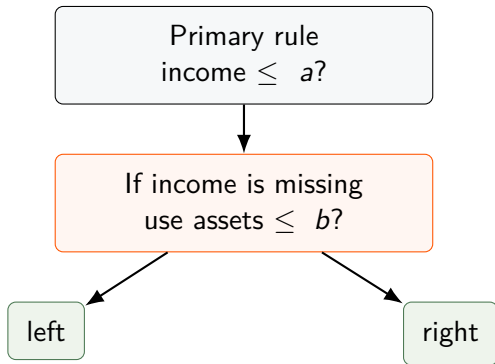
Learn (feature, threshold, missing direction) at each node.

This is a routing rule. Later splits can still separate missing samples.

A Surrogate Split Uses Another Feature



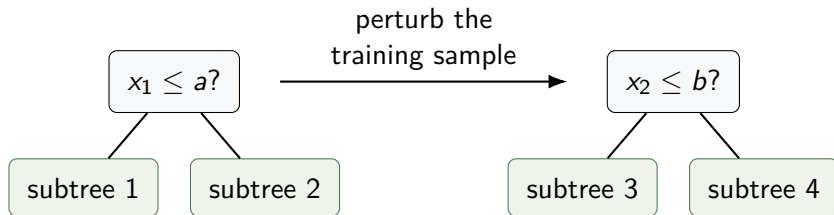
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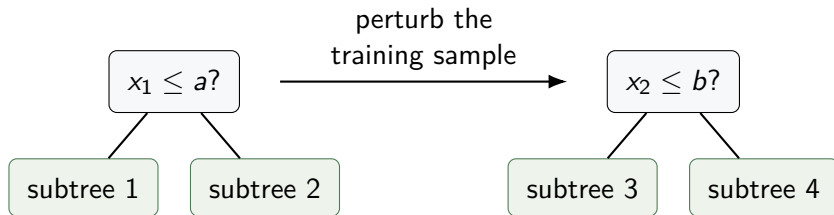
- Find a rule that reproduces the primary split's left/right assignments.
- Samples missing the primary feature may take different branches.
- A learned default direction can also separate samples later, using other features.

CART-style surrogate splits; illustrative rules.

A Small Data Change Can Change the Tree



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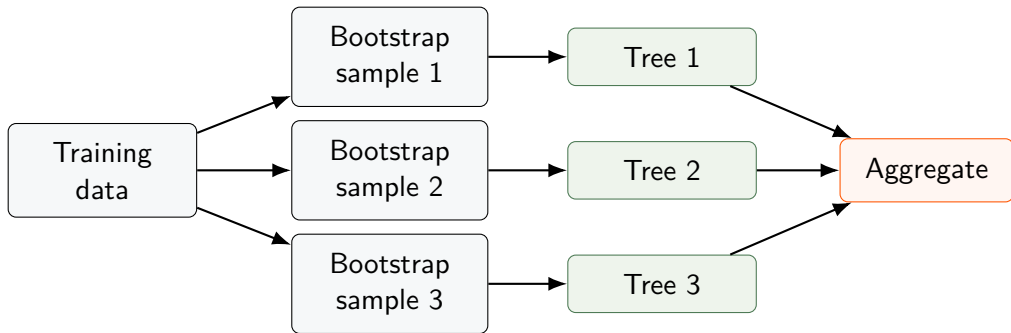


- Competing splits may have very similar training scores.
- A different early split changes the downstream partition.
- Averaging many diverse trees can stabilize predictions.

Schematic illustration of tree instability. Next: random forests.

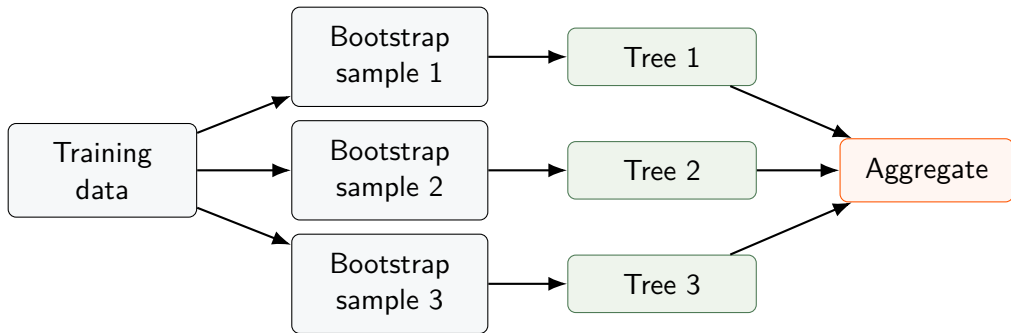
Random Forests

Random Forest: Two Sources of Randomness



- Sample training rows with replacement for each tree.

Random Forest: Two Sources of Randomness



- Sample training rows with replacement for each tree.
- At each node, search a random subset of the features.

Regression: average predictions. Classification: vote or average class probabilities, depending on implementation.

Random Features Reduce Shared Errors

Suppose one feature dominates the split score in many bootstrap samples.

Search every feature

Many trees select similar early splits.

Their prediction errors can be strongly correlated.

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Tradeoff: fewer candidate features can reduce correlation, but may also weaken individual trees.

Correlation Limits the Gain from Averaging

At a fixed input, suppose trees have variance σ^2 and pairwise correlation ρ .

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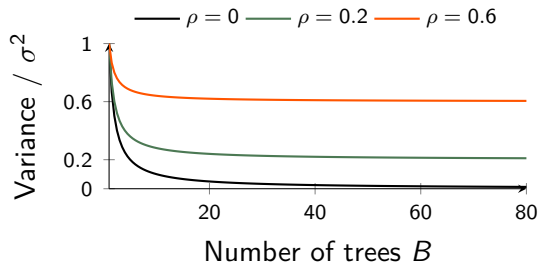
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$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B T_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

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$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B T_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$



More trees reduce the second term.

Lower correlation reduces the limiting variance.

OOB Is Defined Separately for Each Tree

	Tree 1	Tree 2	Tree 3	Tree 4
Sample A	in-bag	OOB	in-bag	OOB
Sample B	OOB	in-bag	in-bag	in-bag
Sample C	in-bag	in-bag	OOB	OOB

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$$\hat{f}_{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} T_b(x_i), \quad \mathcal{B}_i = \{b : i \notin D_b\}, \quad |\mathcal{B}_i| > 0.$$

Compare each sample's OOB prediction with its observed target.

Formula shown for regression; classification can aggregate OOB class probabilities.

OOB Depends on Draws per Tree

Original data: n samples. Each tree draws m times with replacement.

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Draws per tree	OOB probability q	Expected OOB trees, $B = 100$
$m = n$	36.8%	36.8
$m = 2n$	13.5%	13.5
$m = 5n$	0.67%	0.67

Increasing B gives more OOB predictions per sample.

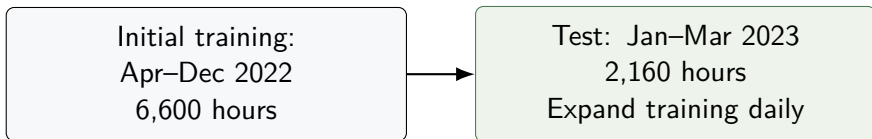
Increasing m/n makes exclusion from any one tree less likely.

Large- n approximations. Standard bootstrap usually takes $m = n$; $m > n$ rows are illustrative variants.

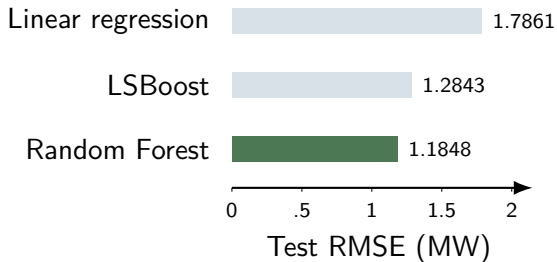
Random Forest for Port Energy Forecasting

Valletta Cruise Port, Malta: a 2025 study

- Predict next-day hourly demand to support energy scheduling.
- Inputs: past demand, calendar, temperature, PV generation, ship activity.
- 8,760 hourly records; metered baseline plus estimated shore-power demand.



Lower Forecast Error Than Linear Regression



33.7%

lower RMSE than linear regression.

All shown models use expanding-window evaluation.

XGboost

Boosting Learns Successive Corrections

Random forest

Trees can be trained independently.

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x).$$

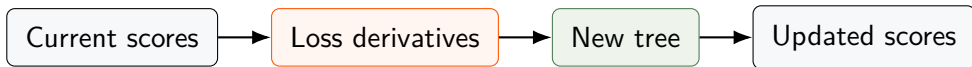
Average diverse predictions.

Gradient boosting

Each new tree depends on the current model.

$$F_t(x) = F_{t-1}(x) + \eta f_t(x).$$

Add a correction to the current score.



Connection to AdaBoost: sequential learning; the loss determines how later learners respond to errors.

Squared Loss Leads to Residual Fitting

Start with $F_0(x) = 5$. Fit a regression stump to $r_i = y_i - F_0(x_i)$.

x_i	1	2	3	4
y_i	1	2	8	9
Residual r_i	-4	-3	3	4

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Stump $f_1(x_i)$	-3.5	-3.5	3.5	3.5

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Residual r_i	-4	-3	3	4
Stump $f_1(x_i)$	-3.5	-3.5	3.5	3.5
$F_1(x_i)$, with $\eta = 0.5$	3.25	3.25	6.75	6.75

$$F_1(x) = 5 + 0.5f_1(x).$$

The next tree fits what remains after this update.

A Tree Outputs a Score, Also for Classification

$$s(x) = b + \sum_{t=1}^T \eta f_t(x), \quad f_t(x) = w_{t,q_t(x)}.$$

Squared-error regression

$$\hat{y} = s,$$

$$\ell(y, s) = \frac{1}{2}(s - y)^2.$$

A leaf changes the predicted value.

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Binary classification: $y \in \{0, 1\}$

$$p = \sigma(s) = \frac{1}{1 + e^{-s}},$$

$$\ell(y, s) = \log(1 + e^s) - ys.$$

A leaf changes the logit; sigmoid gives a probability.

$q_t(x)$ identifies a leaf; $w_{t,v}$ is that leaf's real-valued output.

One Round Optimizes One New Tree

Fix the existing ensemble, with current scores $s_i = F_{t-1}(x_i)$.

$$\mathcal{L}_t(f_t) = \sum_{i=1}^n \ell(y_i, s_i + f_t(x_i)) + \Omega(f_t).$$

$$\Omega(f_t) = \gamma J + \frac{\lambda}{2} \sum_{v=1}^J w_v^2.$$

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- Learn the partition and its J leaf values.
- γ penalizes additional leaves; λ shrinks leaf values.
- First derive an unshrunk tree; apply learning rate η afterward.

L2-regularized scalar-tree formulation.

Approximate the Loss Around the Current Score

For each sample, expand in the proposed correction $f_t(x_i)$:

$$\ell(y_i, s_i + f_t(x_i)) \approx \ell(y_i, s_i) + g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2.$$

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$$g_i = \left. \frac{\partial \ell(y_i, s)}{\partial s} \right|_{s=s_i}, \quad h_i = \left. \frac{\partial^2 \ell(y_i, s)}{\partial s^2} \right|_{s=s_i}.$$

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Drop terms that are constant with respect to the new tree:

$$\tilde{\mathcal{L}}_t = \sum_i \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega(f_t).$$

Exact for squared loss; a local approximation for logistic loss.

The Loss Supplies Two Numbers per Sample

Loss	Gradient g_i	Hessian h_i
$\frac{1}{2}(s_i - y_i)^2$	$s_i - y_i$	1
Binary cross-entropy	$p_i - y_i$	$p_i(1 - p_i)$

In classification, $p_i = \sigma(s_i)$.

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In classification, $p_i = \sigma(s_i)$.

Compute g_i, h_i from the existing ensemble once per round.

Keep them fixed while growing this round's tree.

Candidate leaves only need sums of these statistics.

Each Leaf Has an Optimal Constant Correction

For samples I_v reaching leaf v , define

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$$Q_v(w_v) = G_v w_v + \frac{1}{2}(H_v + \lambda)w_v^2 + \gamma.$$

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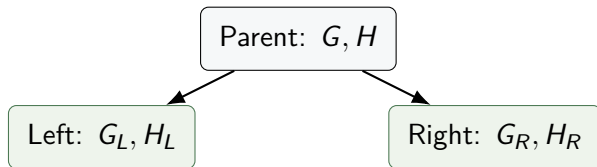
$$Q_v(w_v) = G_v w_v + \frac{1}{2}(H_v + \lambda)w_v^2 + \gamma.$$

$$\frac{\partial Q_v}{\partial w_v} = G_v + (H_v + \lambda)w_v = 0 \quad \implies \quad \boxed{w_v^* = -\frac{G_v}{H_v + \lambda}}.$$

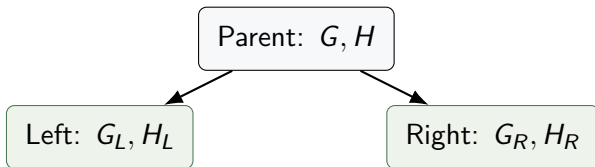
For squared loss and $\lambda = 0$, this is the mean residual in the leaf.

A leaf value is a score correction contributed by this tree.

Split Gain Chooses the Tree Structure

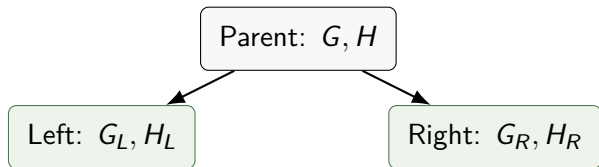


Split Gain Chooses the Tree Structure



$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right] - \gamma$$

Split Gain Chooses the Tree Structure



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- Compare the optimized parent leaf with two optimized child leaves.
- Search candidate features and thresholds; prefer the largest allowed gain.
- Recurse, with growth constraints and pruning controlling complexity.

Here $G = G_L + G_R$, $H = H_L + H_R$. The displayed gain already subtracts γ .

Worked Example: Start a Regularized Round

Restart from $s_i = 5$ for all samples. Use squared loss, $\lambda = 1$, $\gamma = 0$.

Feature x_i	Target y_i	Score s_i	$g_i = s_i - y_i$	h_i
1	1	5		
2	2	5		
3	8	5		
4	9	5		

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Feature x_i	Target y_i	Score s_i	$g_i = s_i - y_i$	h_i
1	1	5	4	1
2	2	5	3	1
3	8	5	-3	1
4	9	5	-4	1

$$G = 0, \quad H = 4.$$

Candidate thresholds: 1.5, 2.5, and 3.5.

Compare the Candidates and Set Leaf Values

For $x \leq 2.5$: $G_L = 7$, $H_L = 2$, $G_R = -7$, $H_R = 2$.

$$\text{Gain} = \frac{1}{2} \left[\frac{49}{3} + \frac{49}{3} - \frac{0}{5} \right] = \frac{49}{3} \approx 16.33.$$

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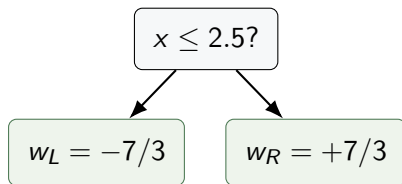
Threshold	Gain
1.5	6
2.5	16.33
3.5	6

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3.5	6



The split groups samples that need similar corrections.

Shrink the New Tree, Then Update the Scores

$$F_t(x) = F_{t-1}(x) + \eta f_t(x).$$

With $\eta = 0.3$:	x_i	1	2	3	4
	Old score	5	5	5	5
	New tree $f_t(x_i)$	$-7/3$	$-7/3$	$7/3$	$7/3$
	Target	1	2	8	9

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With $\eta = 0.3$:	Old score	5	5	5	5
	New tree $f_t(x_i)$	$-7/3$	$-7/3$	$7/3$	$7/3$
	Applied correction	-0.7	-0.7	0.7	0.7
	New score	4.3	4.3	5.7	5.7
	Target	1	2	8	9

The next round recomputes gradients from these updated scores.

In Classification, Leaves Correct the Logit

A leaf contains labels $(1, 1, 1, 0)$; currently $s_i = 0$, so $p_i = 0.5$.

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$$G_v = (-0.5) + (-0.5) + (-0.5) + 0.5 = -1, \quad H_v = 4(0.25) = 1.$$

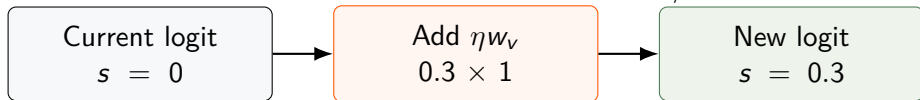
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With $\lambda = 0$, the leaf value is $w_v = -G_v/H_v = 1$.



$$p_{\text{new}} = \sigma(0.3) \approx 0.574.$$

Sum tree outputs first; apply sigmoid to the ensemble score.

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2. Start a tree at the root; aggregate G, H for its samples.
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4. Continue growing subject to constraints; set final leaf values.
5. Update $s_i \leftarrow s_i + \eta w_{q_t(x_i)}$.

Recompute derivatives for the next tree.
Keep the existing trees fixed in this additive training scheme.

Regularization Acts at Several Levels

Control	What it changes
Maximum depth	Limits the complexity of interactions in one tree.
<code>min_child_weight</code>	Requires enough Hessian mass H in each child.
λ and γ	Shrink leaf values and penalize extra leaves.
Learning rate η	Shrinks each tree's contribution to the ensemble.
Row/feature subsampling	Adds randomness during tree construction.

Select boosting rounds on validation data.

Reserve the test set for final evaluation.

XGBoost for Early Sepsis Prediction

A 2023 study: predict sepsis within the next six hours.

- 4,603 trauma patients in MIMIC-IV; one hospital, 2008–2019.
- Clinical variables, measurement patterns, and time-series features.
- Compare XGBoost with L2-regularized logistic regression.

70% development
stratified 5-fold CV

15%
calibration

15%
held-out test

Random patient-level split: each patient's records stay together.

XGBoost vs. Logistic Regression

Six-hour prediction window; time-step-level evaluation.

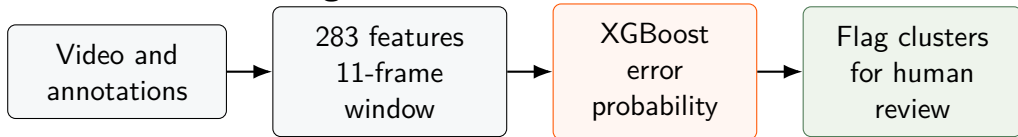
Model	AUROC (95% CI)	AUPRC (95% CI)
L2 logistic regression	0.83 (0.81, 0.85)	0.18 (0.15, 0.21)
XGBoost	0.87 (0.85, 0.89)	0.27 (0.23, 0.31)

AUROC: area under the ROC curve.

AUPRC: area under the precision–recall curve. Higher is better.

XGBoost in Production: Uber, 2026

Detect bounding-box annotation errors before submission.



- Visual, motion, and geometric features summarize video context.
- Training anomalies simulate tracking jumps and identity swaps.
- Deployed across Uber's bounding-box annotation projects.

Three Models, Three Training Strategies

	Single tree	Random forest	XGBoost
Training	Greedy splits	Bootstrap + random features	Sequential loss-based corrections
Combine	One partition	Average or vote	Sum of score increments
Main benefit	Simple rules	Lower variance	Flexible loss minimization
Main controls	Depth, leaf size	Feature subset, leaf size, trees	Depth, regularization, learning rate, rounds

Choose using an evaluation that matches the intended deployment.